

7. GOVERNANCE & RELEASE GATE · STPA Hazard Analysis · Bumpless Human Override · Release Case ('LOOP-01' → 'REL-01')

1. SENSING

Photons / Voltages to DMA · MIPI/SPI

2. PERCEPTION

ViT Encoders · DINOv2 · 3D Tokens

3. MEMORY

Temporal Belief · $SE(3)$ Trees · JEPAs

5. PLANNING (System 1.5 · MPU/NPU)

Diffusion Policies · ACT Chunking (H -steps) · Jerk Continuity

4. REASONING (System 2 · MPU)

Vision-Language Models · 3D Goals · Leases (t_{expire})

----- THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY -----

6. REFLEX (System 1 · MCU)

1 kHz Real-Time Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$

THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics